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Chemistry and Biology of Medicine

The KRAS^{G12C} Story

Introduction

KRAS is one of the most frequently mutated oncogenes in human cancer and plays a central role in regulating cellular proliferation and differentiation through the MAPK signaling pathway. Mutations in KRAS are especially common in pancreatic, colorectal, and non-small-cell lung cancers (NSCLC), where they drive constitutive downstream signaling and uncontrolled tumor growth. Among these mutations, KRAS^{G12C} is significant because the substitution of glycine with cysteine at codon 12 creates a unique nucleophilic residue that can be selectively targeted through covalent inhibition. As illustrated in Figure 1, KRAS^{G12C} normally cycles between an active GTP-bound state that promotes downstream signaling and an inactive GDP-bound state in which signaling is suppressed. Covalent KRAS^{G12C} inhibitors such as sotorasib and adagrasib selectively bind the inactive GDP-bound conformation through the switch-II pocket (S-IIP), which locks KRAS in its inactive state and prevents tumor proliferation.

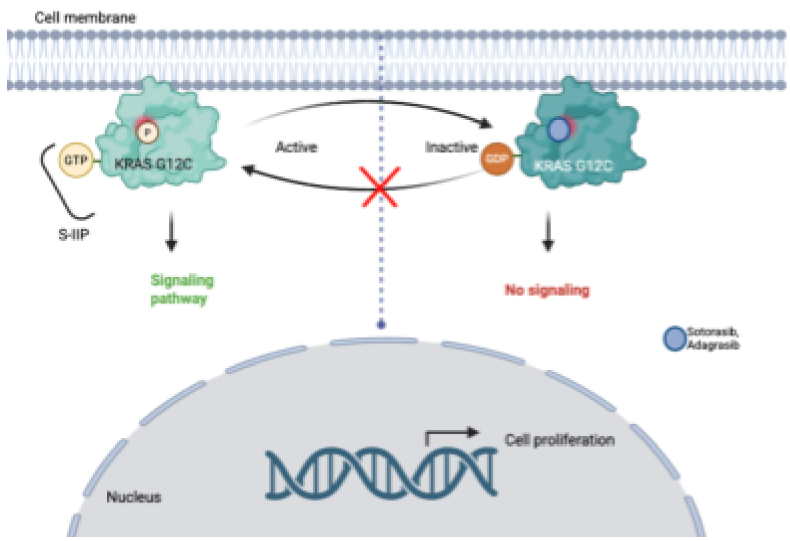

Figure 1: The KRAS^{G12C} GTPase cycle between active GTP-bound and inactive GDP-bound states, illustrating the mechanism of covalent inhibition via the S-IIP.

Despite its importance in cancer biology, KRAS was considered “undruggable” for decades because of its extremely high affinity for GTP/GDP and the apparent absence of a druggable binding pocket. This changed with the discovery of the S-IIP by Ostrem et al. (2013), a transient allosteric cavity accessible only in the inactive GDP-bound form of KRAS^{G12C}. This discovery established the structural foundation for a new generation of mutant-selective covalent inhibitors capable of irreversibly targeting Cys12 and stabilizing KRAS in its inactive state. Early compounds such as Compound 12 demonstrated the feasibility of selective KRAS^{G12C} inhibition, while later inhibitors including ARS-853 and ARS-1620 progressively improved cellular engagement, pharmacokinetic stability, and in vivo efficacy. These advances ultimately led to the development of clinically successful inhibitors such as AMG 510 (sotorasib) and MRTX849 (adagrasib), which became the first KRAS-targeted therapies to produce objective tumor regressions in human patients as demonstrated in Figure 2.

However, despite these major breakthroughs, acquired resistance rapidly emerged as a significant clinical challenge. Tumors frequently restore MAPK signaling through adaptive ERK reactivation. More recent studies involving the dual prenyltransferase inhibitor FGTI-2734 demonstrated that blocking WT RAS membrane localization can synergistically enhance KRAS^{G12C} inhibition. This paper will therefore examine the structural evolution, mechanistic development, clinical translation, and emerging resistance pathways of KRAS^{G12C} inhibitors.

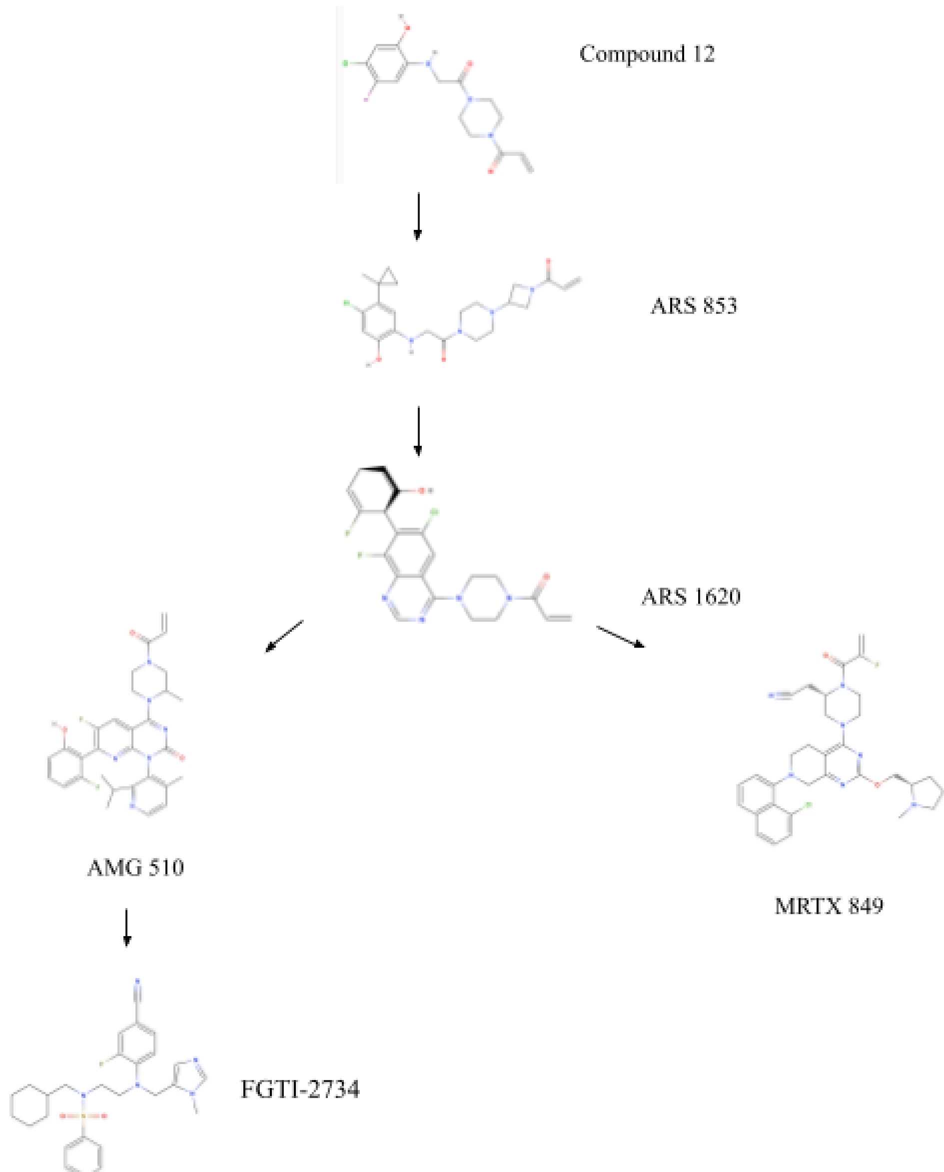

Figure 2: Structural evolution of KRAS^{G12C} inhibitors from Compound 12 to the clinically approved inhibitors sotorasib and adagrasib, with FGTI-2734 as a resistance strategy.

Paper one: KRAS^{G12C} inhibitors allosterically control GTP affinity and effector interactions

The foundational breakthrough in KRAS^{G12C} drug discovery came from Ostrem et al. (2013), who were the first to identify a druggable allosteric pocket in KRAS which they named Switch II Pocket (S-IIP). The S-IIP becomes accessible only when KRAS^{G12C} is in its inactive GDP-bound state. The major objectives of the study were therefore to identify small molecules capable of selectively targeting the G12C mutation and determine whether KRAS could be inhibited allosterically..

To identify an initial chemical starting point, the authors used disulfide-fragment tethering combined with electrospray ionisation mass spectrometry (ESI-MS), a fragment-based drug discovery strategy that exploited the unique nucleophilicity of the mutant cysteine at codon 12. KRAS^{G12C} protein was first loaded into its inactive GDP-bound state and incubated with a library of 480 disulfide-containing fragments. Because each fragment could reversibly form a disulfide bond with a free cysteine residue, β-mercaptoethanol (βME) was added simultaneously as a competing thiol. This ensured that only fragments with genuine non-covalent affinity for the local environment surrounding Cys12 would remain bound. After incubation, ESI-MS was used to quantify the proportion of KRAS molecules modified by each fragment.

The results of this screen revealed two major hit fragments, 6H05 and 2E07, shown in Figure 3.

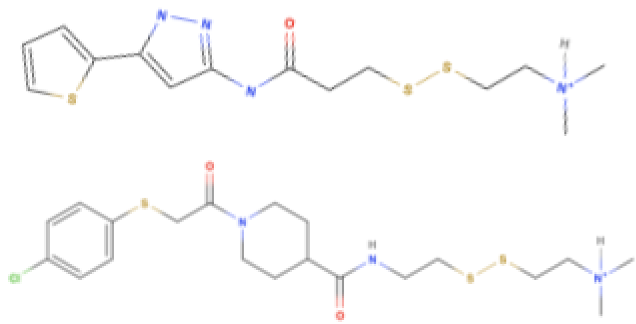

Figure 3: Molecular structures of the two lead fragment hits, 2E07 (top) and 6H05 (bottom), identified by disulfide-fragment tethering screening.

As illustrated in Figure 4, both fragments achieved ~ 85–95% modification of $KRAS^{G12C}$ while producing almost no detectable modification of wild-type (WT) KRAS, despite it having three native cysteine residues (Cys51, Cys80, and Cys118). This demonstrated that fragment binding depended on the unique structural environment surrounding mutant Cys12. Importantly, both fragments also efficiently modified $HRAS^{G12C}$, a closely related RAS paralogue. Structural analysis further revealed that the Switch II Pocket is accessible only in the GDP-bound inactive conformation. In the active GTP-bound state, the Switch II region adopts a different conformation that closes the pocket entirely.

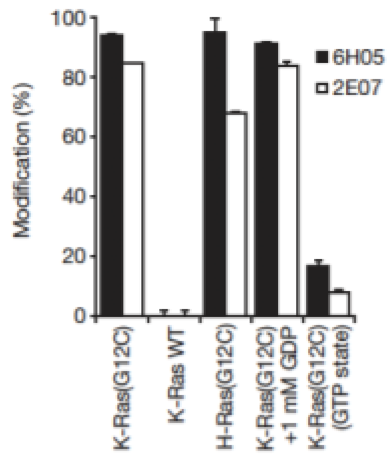

Figure 4: Selective modification of KRAS^{G12C} by fragments 6H05 and 2E07, with minimal modification of wild-type KRAS or the GTP-bound conformation.

Following identification of the fragment hits and discovery of the allosteric pocket, the authors transitioned from reversible disulfide fragments to irreversible carbon-based electrophiles, including acrylamides and vinyl sulphonamides, capable of forming permanent covalent bonds with Cys12. Nearly 100 analogues were synthesized using co-crystal structural guidance to optimize interactions within the S-IIP. This optimization revealed that the hydrophobic dichlorophenyl group formed several critical interactions within the pocket, while residues Gly60 and Glu99 contributed stabilizing hydrogen bonds to the inhibitor. The most effective compound generated through this process was Compound 12, shown in Figure 5, which achieved nearly 100% KRAS^{G12C} modification at 10 µM over 24 hours.

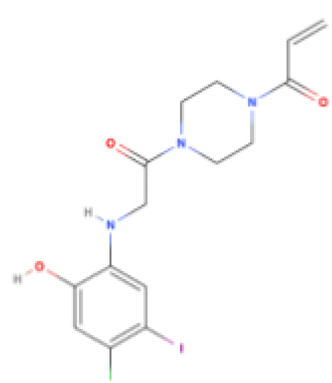

Figure 5: Full structure of Compound 12
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To determine whether these biochemical effects translated into selective antitumor activity, Compound 12 was tested across a panel of lung cancer cell lines using the CellTiter-Glo luminescent viability assay. Cells were treated with increasing concentrations of Compound 12 for 72 hours, and viability was quantified by ATP-dependent luminescence as a measure of live cell number. As shown in Figure 6, all four $KRAS^{G12C}$ cell lines demonstrated dose-dependent decreases in viability, with H1792 cells showing the greatest sensitivity. In contrast, the three non-G12C cell lines showed essentially no response across the same concentration range, which demonstrates the antiproliferative activity of Compound 12.

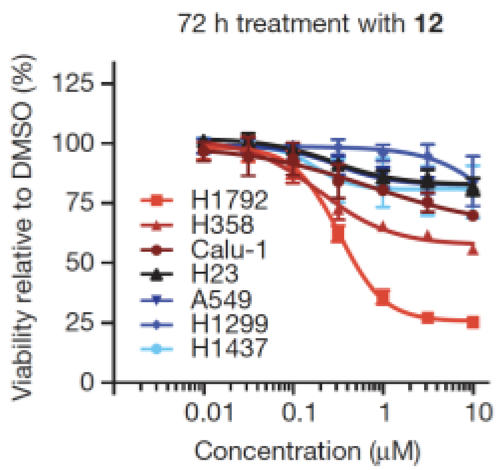

Figure 6: Dose-response viability curves showing selective antiproliferative activity of Compound 12 in $KRAS^{G12C}$ cell lines versus non-G12C controls

Overall, Ostrem et al. (2013) provided the first proof that $KRAS^{G12C}$ could be selectively inhibited through a druggable allosteric pocket. Although Compound 12 itself possessed weak cellular potency and poor drug-like pharmacokinetic properties, the discovery of the S-IIP transformed the KRAS field. This structural finding became the foundation for every subsequent $KRAS^{G12C}$ inhibitor developed over the following decade, including ARS-853, ARS-1620, AMG 510, and adagrasib.

Paper 2: Selective Inhibition of Oncogenic KRAS Output with Small Molecules Targeting the Inactive State

Following the discovery of Compound 12 and the identification of the S-IIP, Patricelli et al. (2016) sought to overcome one of the major limitations of the early fragment inhibitors: the inability to achieve meaningful target living cells. Although Compound 12 could covalently bind KRAS^{G12C} in vitro, it failed to demonstrate measurable cellular activity. To address this, the authors developed a new LC/MS-MS-based cellular engagement assay capable of quantitatively measuring whether a compound forms a covalent bond with Cys12 within intact cells. Using this assay, they confirmed that Compound 12 produced essentially no detectable cellular engagement, revealing that the major issue was not biochemical potency.

To improve cellular activity, the authors synthesized and optimized a series of analogues. An early intermediate, ARS-107, was the first compound to demonstrate promising cellular engagement, and subsequent structure–activity relationship studies identified the 5-chloro position on the phenyl ring as particularly important for activity. As shown in Figure 7, optimization then focused on two major structural modifications, first linker modification to improve the trajectory and positioning of the electrophilic warhead, and then hydrophobic binding pocket modification.

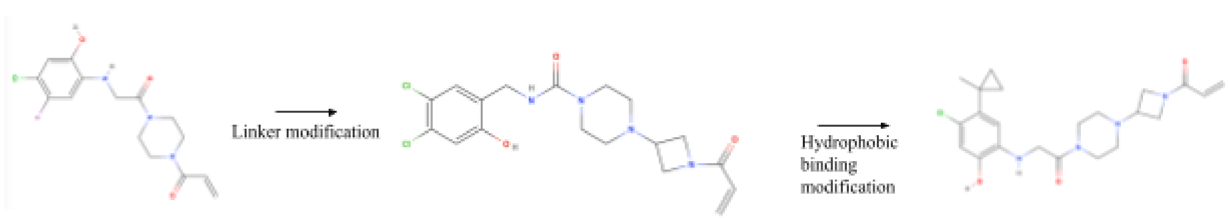

Figure 7: Structural optimization pathway from Compound 12 to ARS-853 via linker modification and hydrophobic binding pocket refinement.

These refinements ultimately led to the development of ARS-853, shown in Figure 8, the first $KRAS^{G12C}$ inhibitor to demonstrate genuine cellular potency.

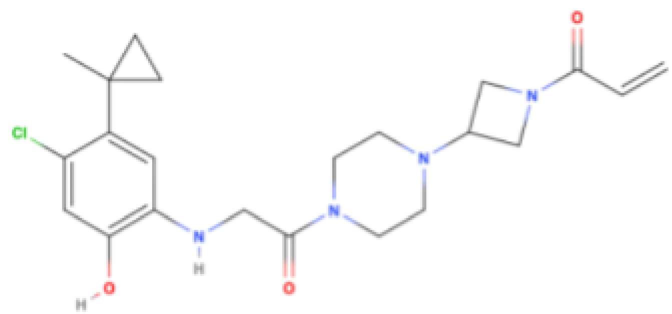

Figure 8: Full structure of ARS-853
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ARS-853 engaged $KRAS^{G12C}$ with a biochemical rate constant of 76 $M^{-1}s^{-1}$ and achieved a six-hour cellular engagement IC_{50} of approximately 1.6 μmol/L. Importantly, this study also challenged the long-standing assumption that mutant KRAS is permanently locked in its active GTP-bound state. Instead, the authors demonstrated that $KRAS^{G12C}$ rapidly cycles between active GTP-bound and inactive GDP-bound conformations in living cells.

The functional consequences of this inhibition are shown in Figure 9. ARS-853 produced dose-dependent suppression of $KRAS^{G12C}$ cancer cell growth while maintaining selectivity for the mutant allele. In standard 2D adherent growth conditions, only a subset of $KRAS^{G12C}$ cell lines showed strong sensitivity, reflecting the known reduction in KRAS dependency under adherent culture conditions. However, under 3D ultra-low adherent conditions that better mimic tumor growth in vivo, all 11 $KRAS^{G12C}$ cell lines demonstrated strong growth inhibition with an average IC_{50} of approximately 2 μmol/L. In contrast, none of the 13 non-G12C mutant cell lines exhibited significant inhibition. Collectively, these findings established ARS-853 as the first direct KRAS inhibitor capable of achieving cellular potency.

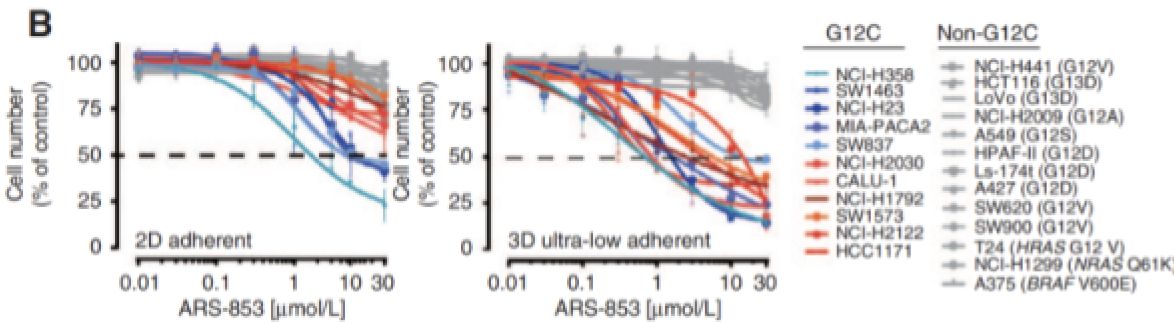

Figure 9: ARS-853 selectively inhibits all 11 KRAS^{G12C}cell lines under 3D ultra-low adherent conditions, with no effect on non-G12C lines.

Despite these major advances, ARS-853 still possessed important limitations that prevented clinical translation. The ortho-amino phenol scaffold produced poor pharmacokinetic properties which directly motivated the development of next-generation inhibitors such as ARS-1620.

Paper 3: Targeting KRAS Mutant Cancers with a Covalent G12C-Specific Inhibitor

Structure-based optimization led to the development of ARS-1620, shown in Figure 10. The original ARS-853 glycine linker and ortho-amino phenol moiety were replaced with a rigid quinazoline core scaffold that more effectively positioned the acrylamide warhead and fluorophenol hydrophobic binding group within the S-IIP. Importantly, ARS-1620 exists as an S-atropisomer, whose axial chirality locks the molecule into a favorable 3D conformation for selective covalent interaction with Cys12. These modifications significantly improved both potency and metabolic stability compared with ARS-853.

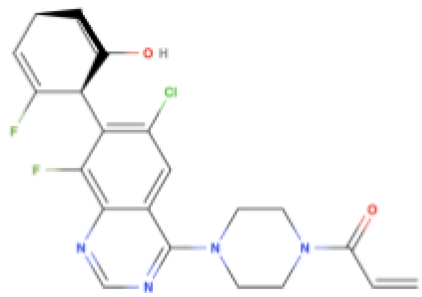

Figure 10: Full structure of ARS 1620
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The structural basis for this improved activity was revealed through X-ray co-crystallography of ARS-1620 bound to KRAS^{G12C}. Figure 11 shows how ARS-1620 occupies the S-IIP while forming stabilizing interactions with nearby amino acid residues, including a previously undescribed interaction with His95. Compared with ARS-853, ARS-1620 filled additional regions of the pocket more effectively, which resulted in approximately a 10-fold improvement in potency. Importantly, ARS-1620 maintained exclusive preference for GDP-bound KRAS^{G12C} and not WT KRAS.

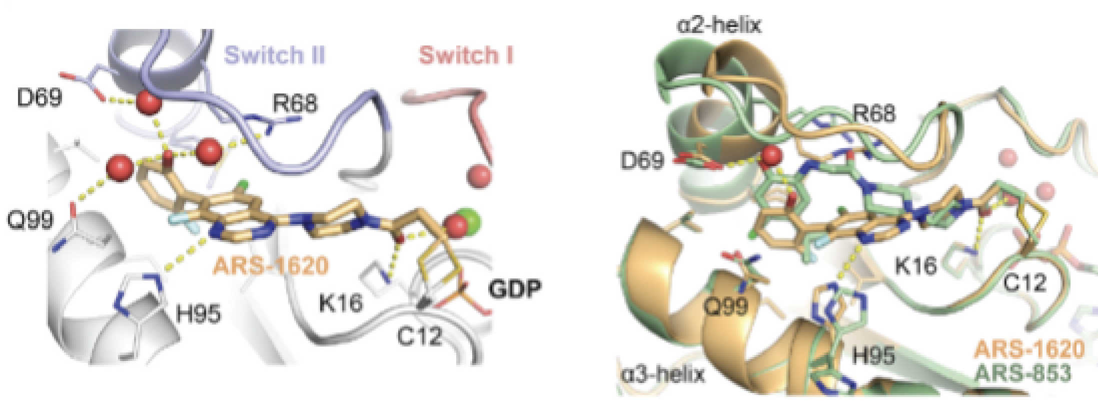

Figure 11: X-ray co-crystallography showing ARS-1620 bound within the S-IIP of KRAS^{G12C}, including a novel interaction with His95, compared with ARS-853.

Together, the findings from Janes et al. (2016) and the 2018 *Cell* study established covalent S-IIP inhibition as a clinically viable therapeutic strategy and laid the foundation for the development of more advanced KRAS^{G12C} inhibitors such as AMG 510 and MRTX849. The biological rationale for these inhibitors is illustrated in Figure 12.

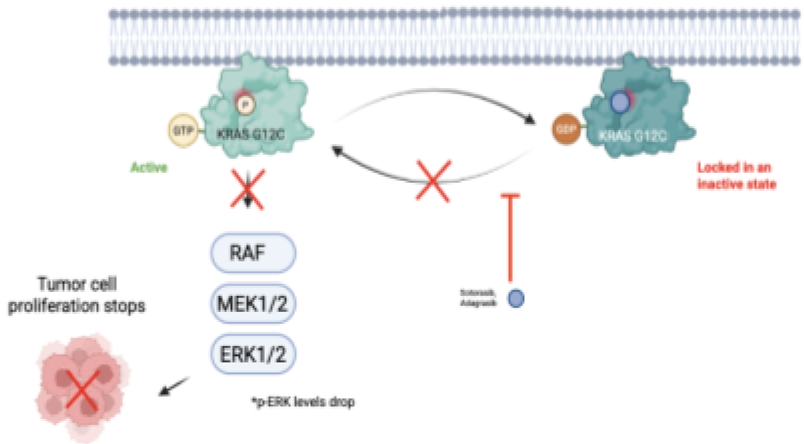

Figure 12: Mechanism of KRAS^{G12C} inhibition of the MAPK pathway, showing suppression of RAF–MEK–ERK signaling and reduced tumor cell proliferation.

In KRAS^{G12C} cancers, KRAS continuously drives MAPK signaling through RAF, MEK, and ERK, promoting uncontrolled proliferation. Covalent inhibitors selectively trap KRAS in its inactive GDP-bound state, which suppresses downstream MAPK signaling, and reduced p-ERK levels, which inhibits tumor growth.

Paper 4: The clinical KRAS^{G12C} inhibitor AMG 510 drives anti-tumour immunity

AMG 510 (sotorasib) represented a major breakthrough in KRAS-targeted therapy by becoming the first clinically successful inhibitor capable of selectively targeting KRAS^{G12C}. AMG 510 was designed as a covalent inhibitor that irreversibly binds to Cys12 within S-IIP, locking KRAS^{G12C} in its inactive GDP-bound state and preventing downstream MAPK signaling. Figure 13 illustrates both the chemical structure of AMG 510 and the X-ray crystallographic interaction between the drug and KRAS^{G12C}. Importantly, the structure also highlights interactions near residues His95 and Gln99, a previously undescribed surface groove that enhances binding efficiency and contributes to its superior kinetic properties. This structural optimization was critical because it allowed AMG 510 to bind mutant KRAS more rapidly and selectively than earlier inhibitors such as ARS-1620.

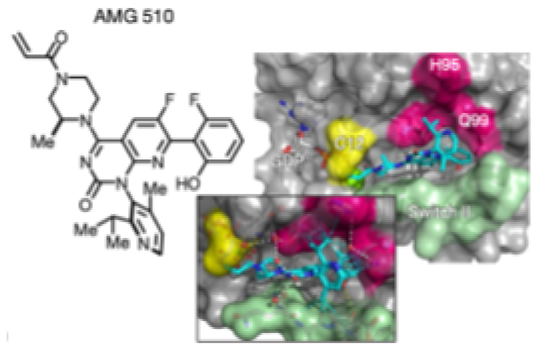

Figure 13: Chemical structure of AMG 510 and X-ray crystallography showing its covalent engagement with KRAS^{G12C} at the S-IIP, highlighting interactions with His95 and Gln99.

The kinetic superiority of AMG 510 was demonstrated using a p-ERK inhibition assay, where suppression of phosphorylated ERK served as a readout of KRAS signaling inhibition. In KRAS^{G12C} NCI-H358 and MIA PaCa-2 cell lines, cells were treated with increasing concentrations of either AMG 510 or ARS-1620, and the observed rate constants (k_{obs}) for pathway inhibition were calculated. Figure 14 shows that AMG 510 consistently achieved substantially higher k_{obs} values across all concentrations tested compared with ARS-1620. This demonstrated that AMG 510 inhibited KRAS^{G12C} signaling much faster than ARS-1620 with a 22-fold increase in kinetic efficiency. This faster covalent engagement translated into quicker inhibition of p-ERK signaling. The experiment therefore established that AMG 510 is a kinetically optimized compound specifically engineered for rapid target engagement.

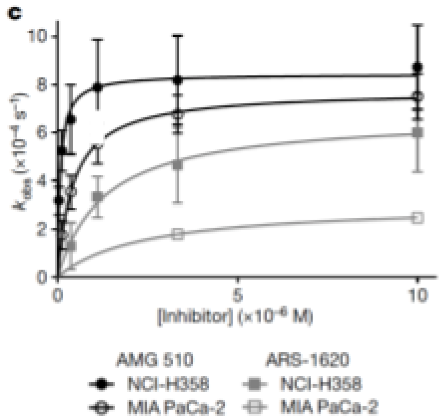

Figure 14: Kinetic superiority of AMG 510 over ARS-1620: observed rate constants (k_{obs}) derived from p-ERK inhibition plotted against inhibitor concentration in NCI-H358 and MIA PaCa-2 cells.

The authors next sought to determine whether covalent binding of AMG 510 directly translated into suppression of KRAS signaling within living tumors. To test this, mice bearing NCI-H358 xenografts received escalating oral doses of AMG 510, and tumors were harvested two hours later for pharmacodynamic analysis. Two parallel measurements were performed on the same tumor tissue. First, mass spectrometry was used to quantify $KRAS^{G12C}$ occupancy by detecting AMG 510–Cys12, expressed as the percentage of total $KRAS^{G12C}$ protein bound by drug. Second, phosphorylated ERK (p-ERK) levels were measured as the downstream signaling readout. Figure 15 reveals the inverse relationship between these two parameters. As AMG 510 dose increased, $KRAS^{G12C}$ occupancy steadily approached nearly 100%, while p-ERK levels simultaneously declined toward near-complete suppression. At the highest dose of 100 mg/kg, the majority of $KRAS^{G12C}$ protein was occupied by AMG 510, and downstream MAPK signaling was almost entirely silenced. Figure 15 therefore provides molecular proof that covalent occupancy of $KRAS^{G12C}$ suppresses oncogenic signaling in living tumors.

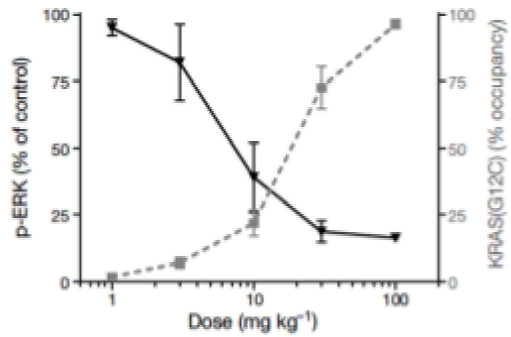

Figure 15: Inverse relationship between KRAS^{G12C} occupancy (dashed, right axis) and p-ERK levels (solid, left axis) in NCI-H358 tumours 2 hours after a single oral dose of AMG 510

AMG 510 was evaluated in a first-in-human phase I dose-escalation trial involving heavily pre-treated patients with KRAS^{G12C} non-small-cell lung cancer (NSCLC). Patients received oral AMG 510 once daily at doses of either 180 mg or 360 mg, and tumor responses were monitored using computed tomography (CT) imaging. Figure 16 presents representative CT scans from two patients before and after treatment. In the left panel, tumors visibly decreased in size after six weeks of therapy at 180 mg, with reductions of approximately 32–50% across multiple lesions. In the right panel, the patient treated at 360 mg showed complete disappearance of some target lesions by 18 weeks. Importantly, these patients had previously failed standard therapies including carboplatin, pemetrexed, and nivolumab.

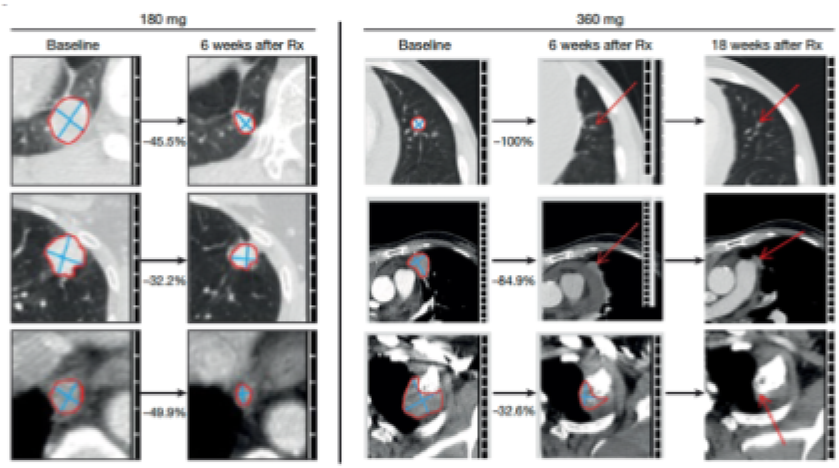

Figure 16: CT scans from the first-in-human trial showing objective partial responses at 180 mg (left, 6 weeks post-treatment) and 360 mg (right, 6 and 18 weeks post-treatment), including complete resolution of target lesions at 18 weeks in the 360 mg cohort.

Collectively, these findings established AMG 510 as the first clinically successful $KRAS^{G12C}$ inhibitor. Structural optimization allowed AMG 510 to achieve rapid covalent engagement and kinetically superior inhibition of $KRAS^{G12C}$. This covalent occupancy directly translated into proportional suppression of downstream MAPK signaling in vivo, and ultimately into objective tumor regression in patients with advanced KRAS-mutant lung cancer.

Paper: Identification of the Clinical Development Candidate MRTX849, a Covalent $KRAS^{G12C}$ Inhibitor for the Treatment of Cancer

During the same period that AMG 510 (sotorasib) was emerging as the first clinically successful $KRAS^{G12C}$ inhibitor, another major $KRAS^{G12C}$ inhibitor was developing in parallel. Mirati Therapeutics identified MRTX849 (later known clinically as adagrasib), a covalent $KRAS^{G12C}$ inhibitor designed to improve potency, and pharmacokinetic stability. The objective of this study is to optimize the tetrahydropyridopyrimidine scaffold to generate a clinically viable inhibitor with improved potency and oral bioavailability, which led to the discovery of MRTX849 as shown in Figure 17.

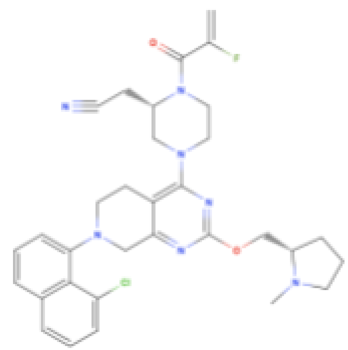

Figure 17: Full structure of MRTX849
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A major concern in the development of covalent inhibitors is selectivity, since irreversible cysteine-reactive drugs can potentially modify many proteins throughout the cellular proteome. To evaluate whether MRTX849 selectively targeted $KRAS^{G12C}$, the authors performed MS-based proteomic cysteine profiling in NCI-H358 cells treated with 3 µM MRTX849 for three hours. Free cysteines were subsequently labeled with a thiol-reactive desthiobiotin probe and analyzed by mass spectrometry. Figure 18 demonstrates the selectivity of MRTX849 across more than 5,700 detected cysteine-containing peptides. Nearly all peptides had minimal engagement by the drug. In striking contrast, $KRAS^{G12C}$ appears as a dramatic outlier with a treated/control ratio near 0.03, corresponding to near-complete occupation of the mutant Cys12 residue. Only one additional protein, KARS, showed modest off-target engagement. This experiment therefore provided strong molecular evidence that MRTX849 covalently and selectively targets $KRAS^{G12C}$ in living cells while leaving the remainder of the cellular cysteine proteome largely unaffected.

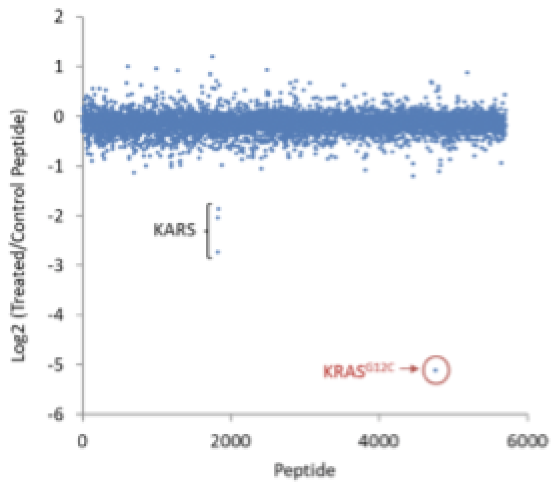

Figure 18: MS-based proteomic cysteine profiling showing near-exclusive engagement of KRAS^{G12C} by MRTX849 across more than 5,700 detected cysteine-containing peptides.

The authors next investigated whether this highly selective target engagement translated into durable antitumor activity in vivo. Female athymic nude mice bearing MIA PaCa-2 pancreatic KRAS^{G12C} xenografts received daily oral doses of MRTX849 at 10, 30, or 100 mg/kg for 16 days. Tumor growth was then monitored over a prolonged 70-day period after treatment initiation. Figure 19 shows that MRTX849 produced dose-dependent tumor regressions. Vehicle-treated tumors grew rapidly and continuously, reaching approximately 1000 mm^3 by day 20. In contrast, all treatment groups initially showed substantial tumor regression. Tumors in the 10 mg/kg group eventually rebounded after treatment cessation. The 30 mg/kg group demonstrated substantially improved durability, with several mice remaining tumor-free through day 70. Most strikingly, mice treated at 100 mg/kg exhibited complete and sustained tumor regression throughout the entire monitoring period. Importantly, no significant body weight loss was observed at any dose level, indicating that the treatment was well tolerated despite prolonged administration.

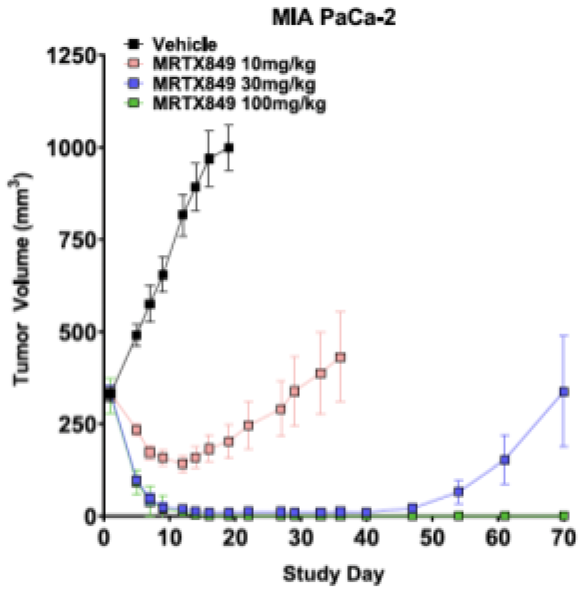

Figure 19: Dose-dependent tumor regression in MIA PaCa-2 xenografts treated with MRTX849 at 10, 30, and 100 mg/kg over a 70-day monitoring period.

Together, these findings established MRTX849 as a potent, selective, and orally bioavailable $KRAS^{G12C}$ inhibitor capable of producing durable complete tumor regressions in vivo.

Paper: FGTI-2734 Inhibits ERK Reactivation to Overcome Sotorasib Resistance in $KRAS^{G12C}$ Lung Cancer

Acquired resistance remains one of the largest limitations of $KRAS^{G12C}$ inhibitors such as sotorasib. Although sotorasib effectively locks $KRAS^{G12C}$ in its inactive GDP-bound state, tumor cells frequently develop mechanisms that restore downstream MAPK/ERK signaling and allow proliferation to continue. Several clinically relevant resistance mechanisms have been identified, including amplification of WT KRAS, amplification or mutation of receptor tyrosine kinases (RTKs) such as EGFR and RET, secondary KRAS mutations, and mutations directly within the S-IIP that prevent sotorasib binding. However, one of the most important and recurring resistance pathways is adaptive ERK reactivation through WT HRAS and NRAS. Even when mutant $KRAS^{G12C}$ is successfully inhibited, tumors can bypass the blockade by activating alternative RAS isoforms that continue signaling through the RAF–MEK–ERK cascade.

To overcome this escape mechanism, Kazi et al. developed FGTI-2734, a dual farnesyltransferase (FT) and geranylgeranyltransferase-1 (GGT-1) inhibitor. The rationale behind this strategy is rooted in RAS membrane biology. WT HRAS and NRAS require lipid modifications, specifically prenylation, to localize to the plasma membrane where signaling occurs. Without membrane localization, these proteins cannot efficiently activate downstream ERK signaling. FGTI-2734 blocks both FT and GGT-1. Figure 20 shows the chemical structure of FGTI-2734, illustrating the small-molecule inhibitor designed to simultaneously target both prenylation enzymes. By targeting a process required for RAS localization rather than a single mutation, FGTI-2734 is capable of suppressing adaptive signaling regardless of the exact resistance mutation involved.

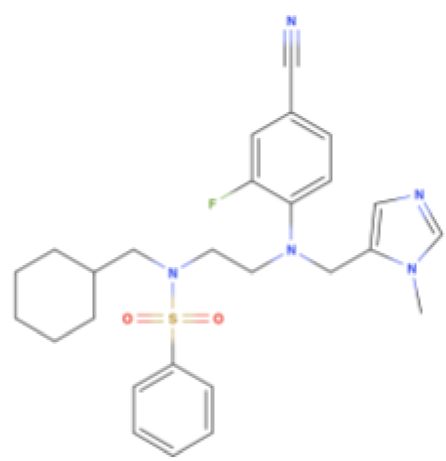

Figure 20: Full structure of FGTI-2734
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The therapeutic rationale for combining FGTI-2734 with sotorasib is therefore highly complementary. Sotorasib directly inhibits mutant $KRAS^{G12C}$, while FGTI-2734 blocks the activation of WT HRAS/NRAS that would otherwise reactivate ERK signaling. This synergistic interaction was demonstrated experimentally using SynergyFinder analysis. In Figure 21, the three-dimensional response surface reveals a pronounced red synergy peak across multiple concentration combinations of sotorasib and FGTI-2734. The red peak visually demonstrates a strong positive synergy score. Importantly, the synergy occurs in a dose-dependent manner, supporting the conclusion that the two drugs interact mechanistically and nonlinearly to suppress tumor cell viability. The accompanying Loewe and Highest Single Agent (HSA) synergy scores further confirm the robustness of this interaction, with consistently positive synergy values observed across models.

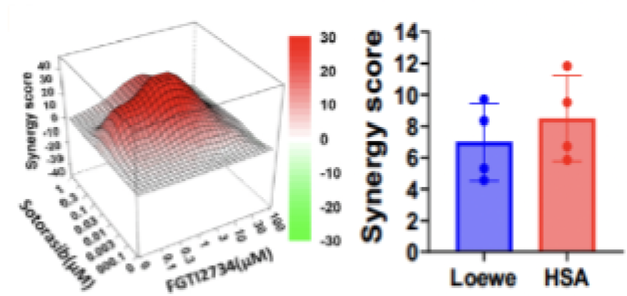

Figure 21: SynergyFinder analysis showing a pronounced synergy peak between sotorasib and FGTI-2734 across multiple concentration combinations, confirmed by Loewe and HSA scores.

The functional consequences of this synergy were further validated through apoptosis assays in highly resistant LU99 cells. Figure 22 shows confocal microscopy images of LU99 cells treated with vehicle, FGTI-2734 alone, sotorasib alone, or the combination. Annexin V-FITC staining (green) was used as a marker of apoptosis, while DAPI staining (blue) labeled cell nuclei. The vehicle-treated cells show almost no green fluorescence, which indicates minimal cell death. Single-agent treatment with either FGTI-2734 or sotorasib induced only modest apoptosis, visible as scattered green staining throughout the panels. In contrast, the combination-treated cells display dramatically increased green fluorescence, indicating a substantial increase in apoptotic cell death. The combination panel contains far more Annexin V-positive cells than either monotherapy condition. These findings were also consistent with biochemical evidence from Western blot analysis showing enhanced Caspase-3 activation and PARP cleavage in the combination-treated cells.

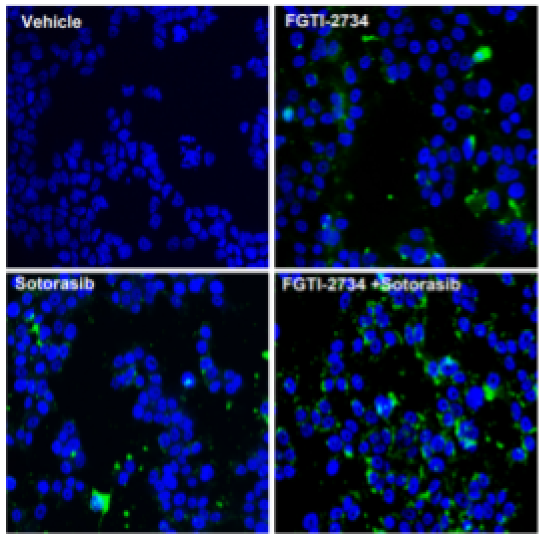

Figure 22: Confocal microscopy of LU99 cells stained with Annexin V-FITC and DAPI, showing dramatically increased apoptosis in cells treated with the FGTI-2734 and sotorasib combination versus either agent alone.

Evidence for the effectiveness of the combination of FGTI-2734 and sotorasib came from the patient-derived xenograft (PDX) model shown in Figure 23. PDX models are considered one of the most clinically relevant preclinical systems because they use real tumor tissue from patients implanted into mice. In this experiment, $KRAS^{G12C}$ lung tumor tissue was implanted into NSG mice and treated with vehicle, sotorasib alone, FGTI-2734 alone, or the combination. Over the 47-day treatment period, the vehicle group showed rapid tumor progression, with tumor volumes increasing dramatically over time. FGTI-2734 alone slowed growth somewhat, while sotorasib alone produced a stronger but still incomplete suppression of tumor progression. By day 47, vehicle-treated tumors had increased by approximately 1333%, FGTI-2734 alone by 563%, and sotorasib alone by 300%, whereas the combination caused approximately 62% tumor regression. Mechanistically, the combination also suppressed phospho-ERK levels in PDX tissue, directly confirming that ERK reactivation was successfully blocked in vivo.

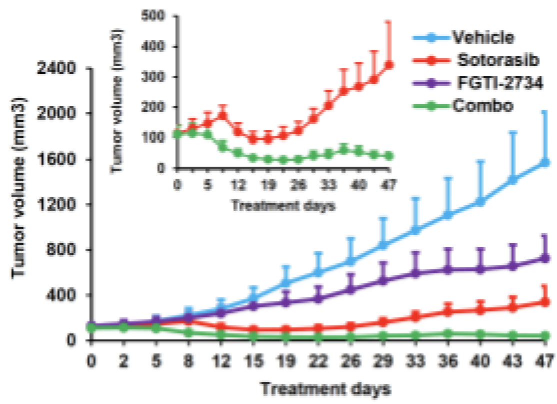

Figure 23: Patient-derived xenograft tumor growth curves over 47 days, demonstrating superior tumor regression with the FGTI-2734 and sotorasib combination compared to either monotherapy.

Overall, this study demonstrates that adaptive ERK reactivation through WT HRAS and NRAS is a major mechanism of resistance to $KRAS^{G12C}$ inhibition. By preventing membrane localization of these compensatory RAS proteins, FGTI-2734 blocks the bypass signaling responsible for resistance and restores sensitivity to sotorasib. The strong synergy observed in viability assays, the dramatic induction of apoptosis in resistant cell lines, and the profound tumor regression seen in patient-derived xenograft models collectively provide compelling evidence that combined $KRAS^{G12C}$ inhibition and RAS prenylation blockade may represent a promising therapeutic strategy for overcoming acquired resistance in KRAS-mutant lung cancer.

Future research

Despite the remarkable success of $KRAS^{G12C}$ inhibitors such as sotorasib and adagrasib, acquired resistance remains one of the greatest challenges limiting durable clinical responses.

The work involving FGTI-2734 demonstrated that adaptive ERK reactivation through WT HRAS and NRAS is a major bypass mechanism that allows tumors to escape $KRAS^{G12C}$ inhibition. Therefore, one important direction for future research would be the development of clinical trials evaluating the combination of FGTI-2734 and sotorasib in human patients. A future phase I clinical trial could initially focus on evaluating safety, tolerability, pharmacokinetics, and dose escalation of the combination in patients with $KRAS^{G12C}$ non-small-cell lung cancer who have progressed on sotorasib monotherapy.

Another promising future direction would involve designing a single bifunctional molecule capable of simultaneously targeting $KRAS^{G12C}$ and preventing WT RAS-mediated bypass signaling. Rather than administering two separate drugs, medicinal chemistry approaches could potentially link the pharmacophores of sotorasib and FGTI-2734 into one optimized compound. Such a molecule would ideally retain the covalent acrylamide warhead required for selective binding to Cys12 within the S-IIP while also preserving the prenyltransferase inhibitory activity necessary to block HRAS and NRAS membrane localization. One possible strategy would involve using a flexible linker to connect the $KRAS^{G12C}$ binding scaffold to the dual FT/GGT-1 inhibitory scaffold.

Conclusion

Over the past decade, $KRAS^{G12C}$ drug discovery has progressed from a seemingly impossible concept to one of the most significant breakthroughs in targeted cancer therapy. This progression began with the discovery of the S-IIP by Ostrem et al. (2013), which demonstrated for the first time that KRAS contained a druggable allosteric site accessible in its inactive

GDP-bound conformation. Although early compounds such as Compound 12 lacked sufficient cellular potency and pharmacokinetic stability, they established the mechanistic foundation for all subsequent $KRAS^{G12C}$ inhibitors.

Subsequent studies progressively optimized this strategy through improvements in cellular engagement, metabolic stability, and pocket occupancy. Patricelli et al. (2016) demonstrated that $KRAS^{G12C}$ is not permanently locked in an active state, but instead cycles dynamically between GDP- and GTP-bound conformations, thereby enabling selective covalent inhibition by ARS-853. Later, ARS-1620 introduced substantial improvements in potency and in vivo efficacy through structure-guided optimization of S-IIP interactions and enhanced stabilization within the binding pocket.

These advances ultimately led to the development of clinically successful inhibitors such as AMG 510 (sotorasib) and MRTX849 (adagrasib). Both compounds demonstrated rapid and selective covalent engagement of $KRAS^{G12C}$, potent suppression of downstream MAPK signaling, and durable tumor regression in vivo. Importantly, AMG 510 became the first KRAS-targeted therapy to produce objective clinical responses in patients with advanced KRAS-mutant cancers. Adagrasib further demonstrated that highly selective $KRAS^{G12C}$ inhibition could achieve antitumor responses with minimal off-target proteomic reactivity.

Despite these major successes, acquired resistance remains a substantial clinical challenge. Tumors frequently restore ERK signaling through compensatory activation of WT HRAS and NRAS, secondary KRAS mutations, or receptor tyrosine kinase signaling. The development of FGTI-2734 demonstrated that simultaneously blocking RAS prenylation and $KRAS^{G12C}$ signaling can overcome adaptive ERK reactivation and restore sensitivity to sotorasib.

Overall, the KRAS^{G12C} field has demonstrated that even the most challenging oncogenic targets can become therapeutically attacked through iterative structural optimization. Future progress will likely focus on overcoming resistance through multi-target therapeutic approaches, improved combination strategies, and next-generation inhibitors capable of producing more durable suppression of oncogenic signaling in KRAS-mutant cancers.

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